



3.1 Coordinate geometry and Voronoi diagrams

Geometry and trigonometry · IB Diploma · Concept guide and original exam-style practice

Understand the concept

The distance between two points and the midpoint of a segment are found from coordinates; the perpendicular bisector of a segment consists of all points equidistant from its two endpoints, found using the midpoint and the negative reciprocal of the segment's gradient. A Voronoi diagram partitions the plane into regions, each containing the points closer to one given site than to any other; region boundaries are perpendicular bisectors between neighbouring sites, and vertices (where ≥ 3 regions meet) are equidistant from those sites — useful for nearest-neighbour problems such as siting a facility away from existing points.

Worked example: follow the reasoning

Two towns are located at $A(2,3)$ and $B(8,7)$ (km). Find the equation of the perpendicular bisector of AB , which forms part of the Voronoi boundary between the regions closer to each town.
Solution: Midpoint = $(5,5)$. Gradient of $AB = (7-3)/(8-2)=2/3$, so the perpendicular gradient is $-3/2$. Equation: $y-5=-1.5(x-5) \rightarrow y=-1.5x+12.5$.

Quick check · multiple choice

In a Voronoi diagram, the edges separating neighbouring regions are formed from:

- A The lines joining each pair of sites
- B The perpendicular bisectors of segments joining neighbouring sites
- C Circles centred at each site
- D The medians of the triangle formed by sites

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Three water pumping stations are located at $A(0,0)$, $B(6,0)$, and $C(3,6)$ (km). (a) [6]
Find the equation of the perpendicular bisector of AB . (b) Find the equation of the perpendicular bisector of AC . (c) Hence find the coordinates of the Voronoi vertex equidistant from A , B and C .

Final answer: _____



Answers and explanation

Question 1 [6]

(a) Midpoint $AB=(3,0)$; AB is horizontal, so the perpendicular bisector is the vertical line $x=3$. (b) Midpoint $AC=(1.5,3)$; gradient $AC=2$, so perpendicular gradient $=-0.5$; equation: $y-3=-0.5(x-1.5) \rightarrow y=-0.5x+3.75$. (c) Substituting $x=3$: $y=-1.5+3.75=2.25$. Vertex: $(3, 2.25)$.

Quick check answer

B. Each edge lies on the perpendicular bisector between the two sites whose regions it separates.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.